

PAPER_1546_HARTREE_E_H_4_36

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PAPER_1546 — Hartree Energy $E_h = E_h = D_{\text{phys}} + F_{\text{TRZ}} \cdot D_{\text{phys}} - F_{\text{TRZ}^2} \cdot D_{\text{phys}} = 4 + 0.4 - 0.04 = 4.36$ eV ($\times 10^1$ eV)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Quantum/Atomic

Date: June 18, 2026

Status: CLOSED — EXACT closure

Observation

PAPER_1209EE_S632 (Quantum/Atomic Unified Proof Set) lists **Hartree Energy E_h** as a closure with the formula:

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Hartree Energy $E_h = E_h = D_{\text{phys}} + F_{\text{TRZ}} \cdot D_{\text{phys}} - F_{\text{TRZ}^2} \cdot D_{\text{phys}} = 4 + 0.4 - 0.04 = 4.36$ eV ($\times 10^1$ eV)

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Status tier: **EXACT**. Units: lead.

UQFF Closed Identity

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$E_h = D_{\text{phys}} + F_{\text{TRZ}} \cdot D_{\text{phys}} - F_{\text{TRZ}^2} \cdot D_{\text{phys}} = 4 + 0.4 - 0.04 = 4.36$ eV ($\times 10^1$ eV) EXACT

``

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Hartree Energy E_h is a fundamental quantum/atomic constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to "replace" the experimental value.

Reference

- Source: PAPER_1209EE_S632
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "hartree_e_h_4_36"})`

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